

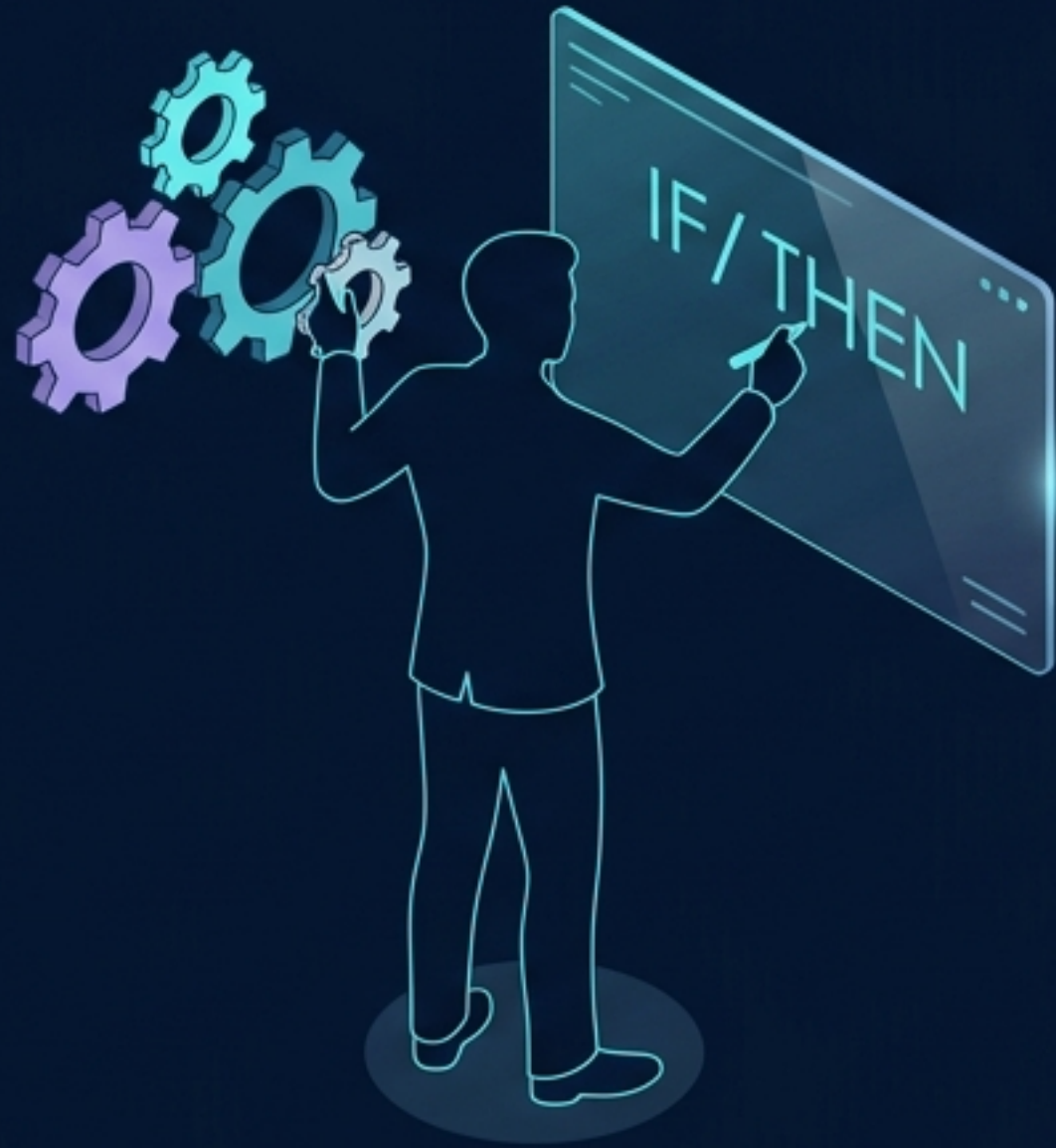
The Logic of Learning: Fundamentals of Machine Learning

Parsa Salamatipour



The Shift – Classic AI vs. Machine Learning

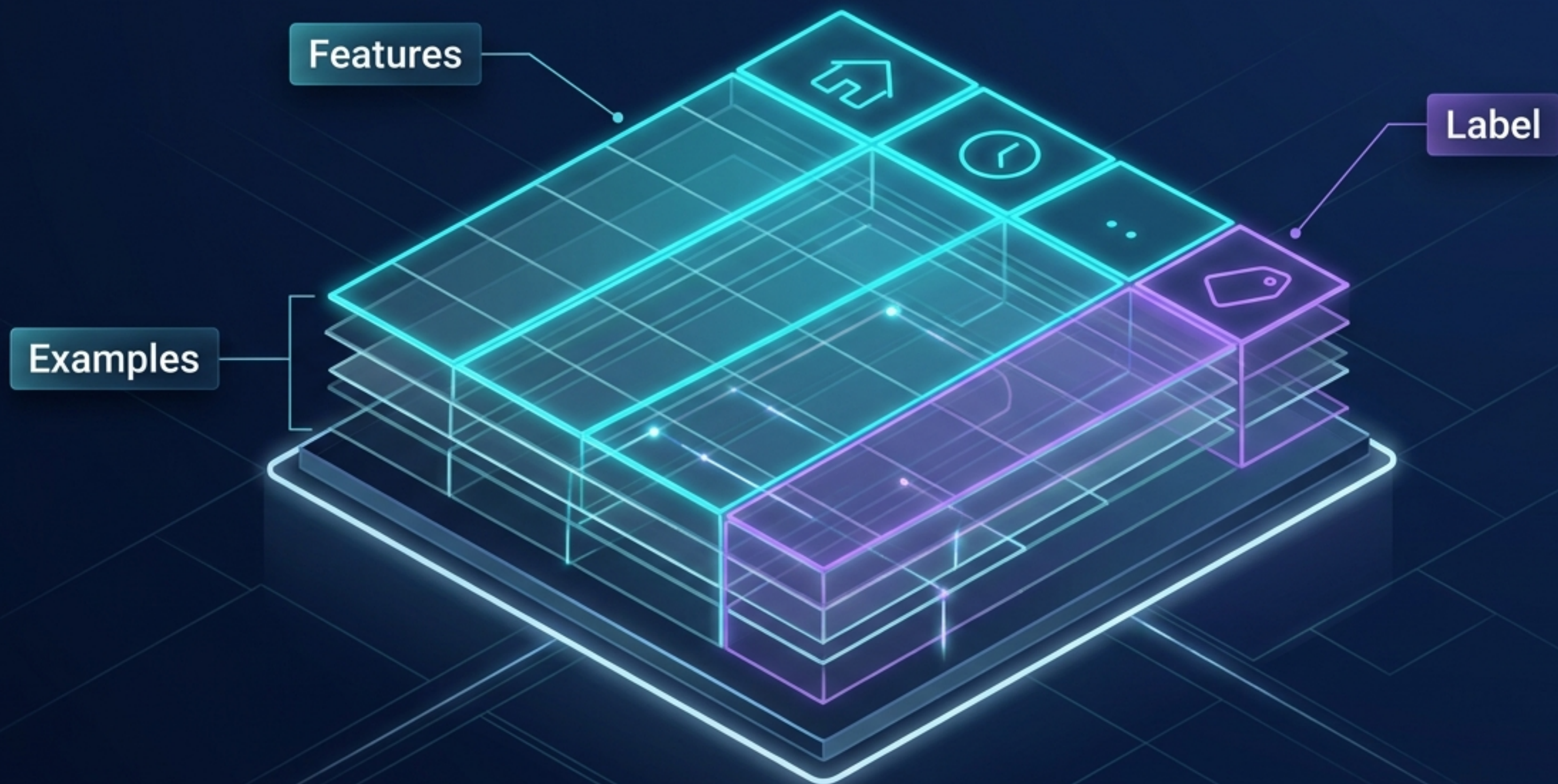
Classic AI



Machine Learning



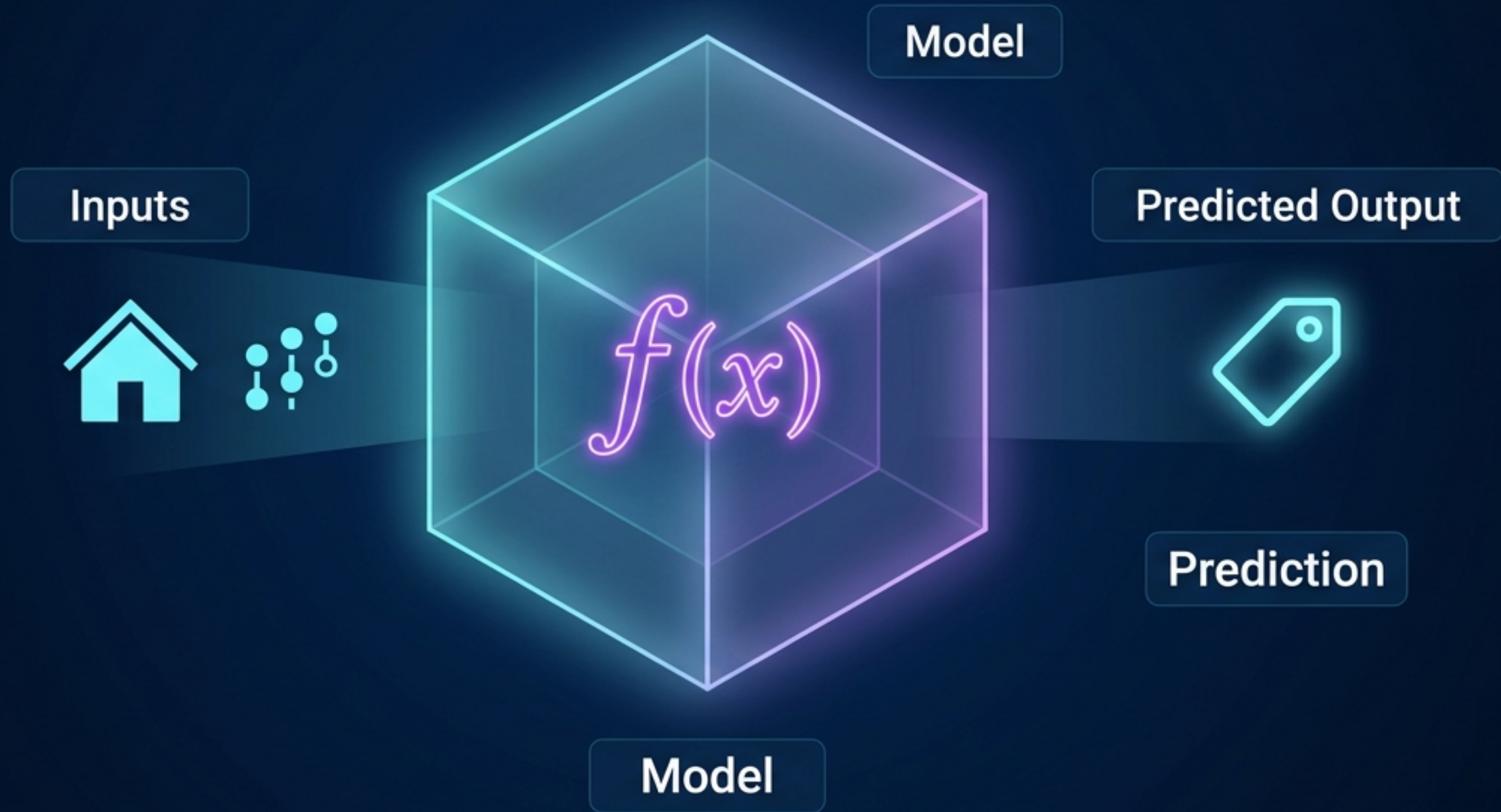
The Anatomy of a Dataset



Which are the Features? Which is the Label?

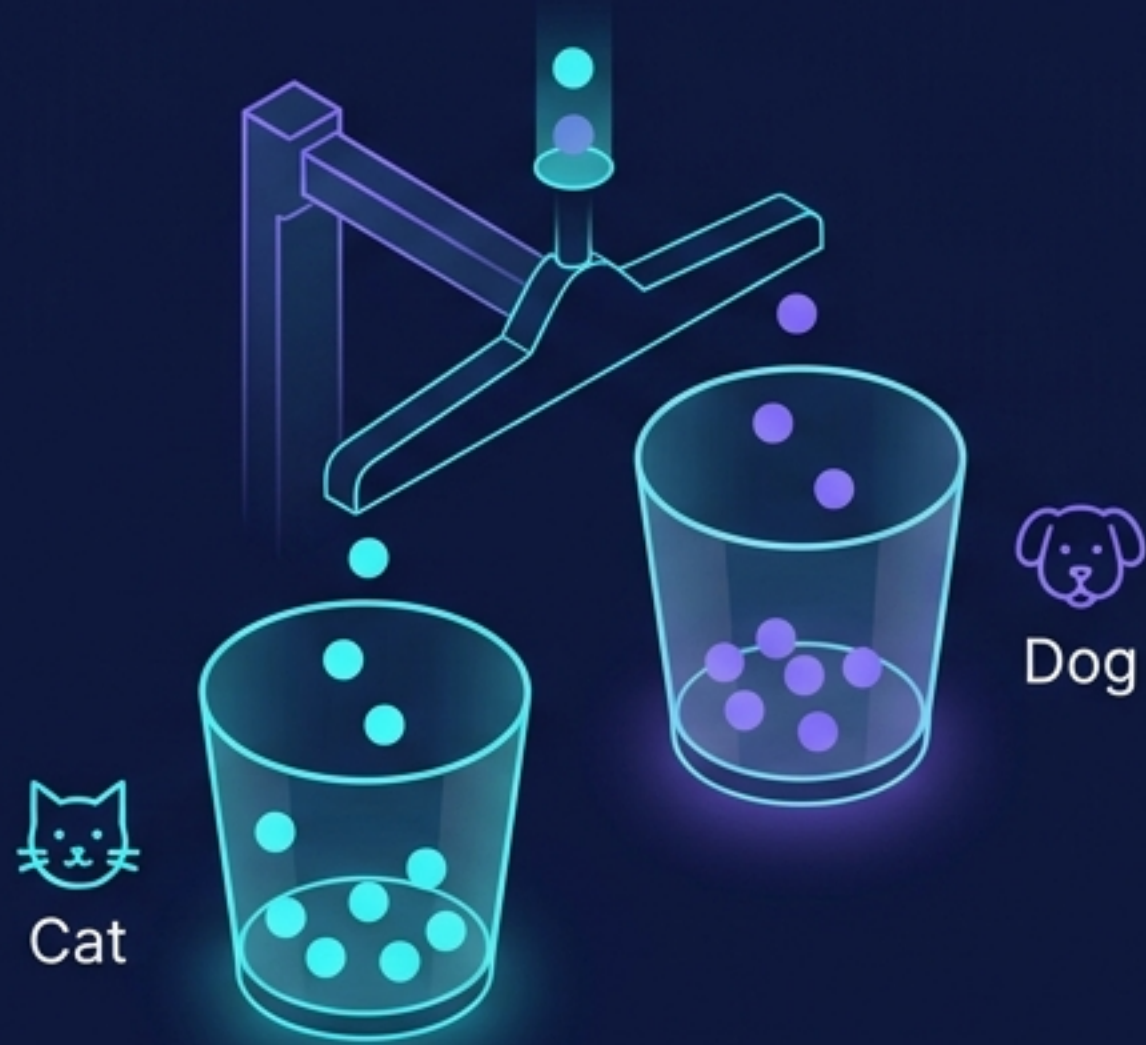


The Black Box – ML as Function Prediction

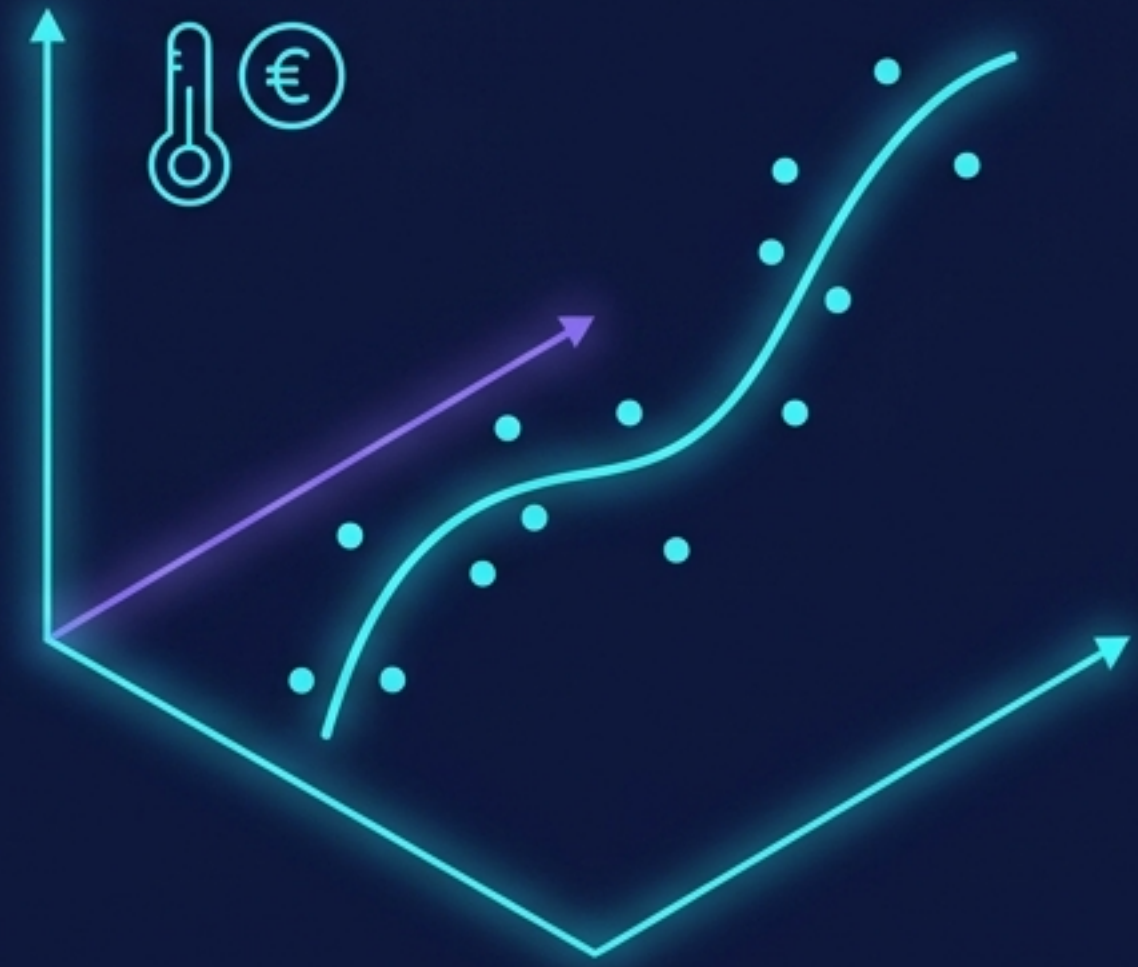


Classification vs. Regression

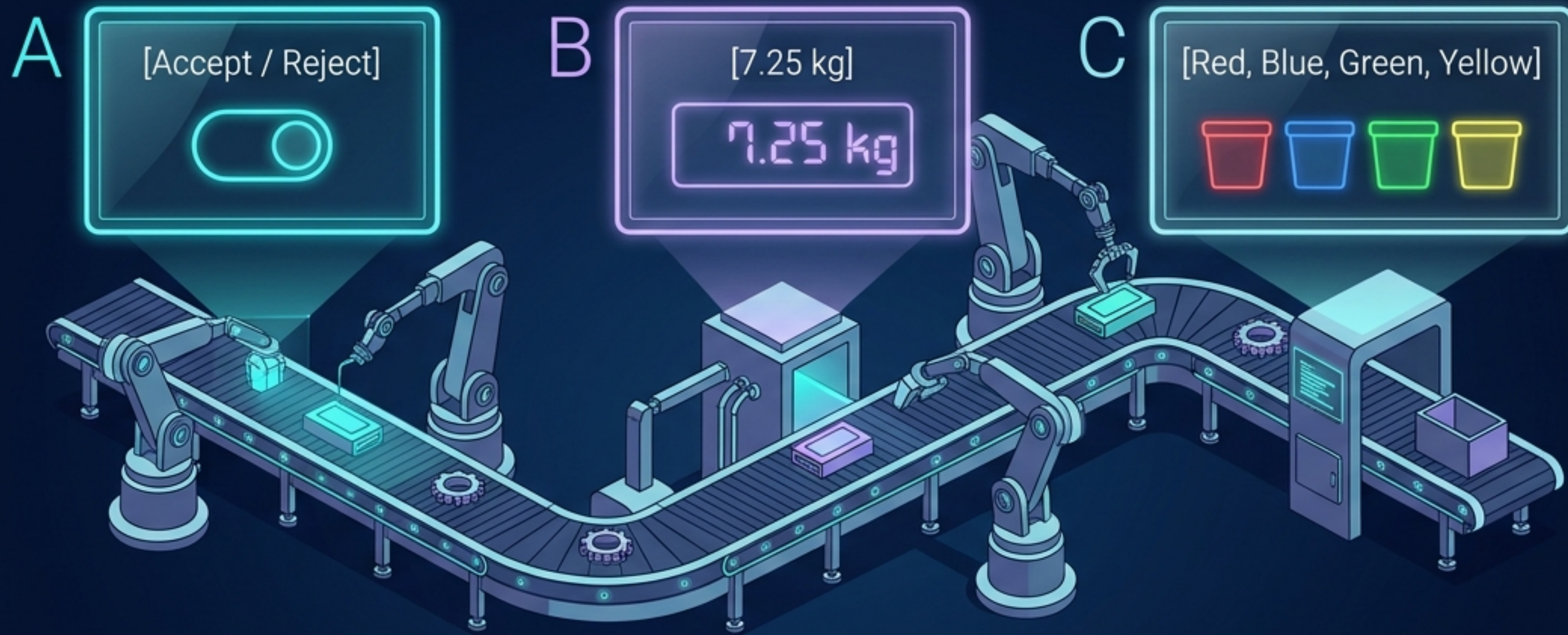
Classification



Regression

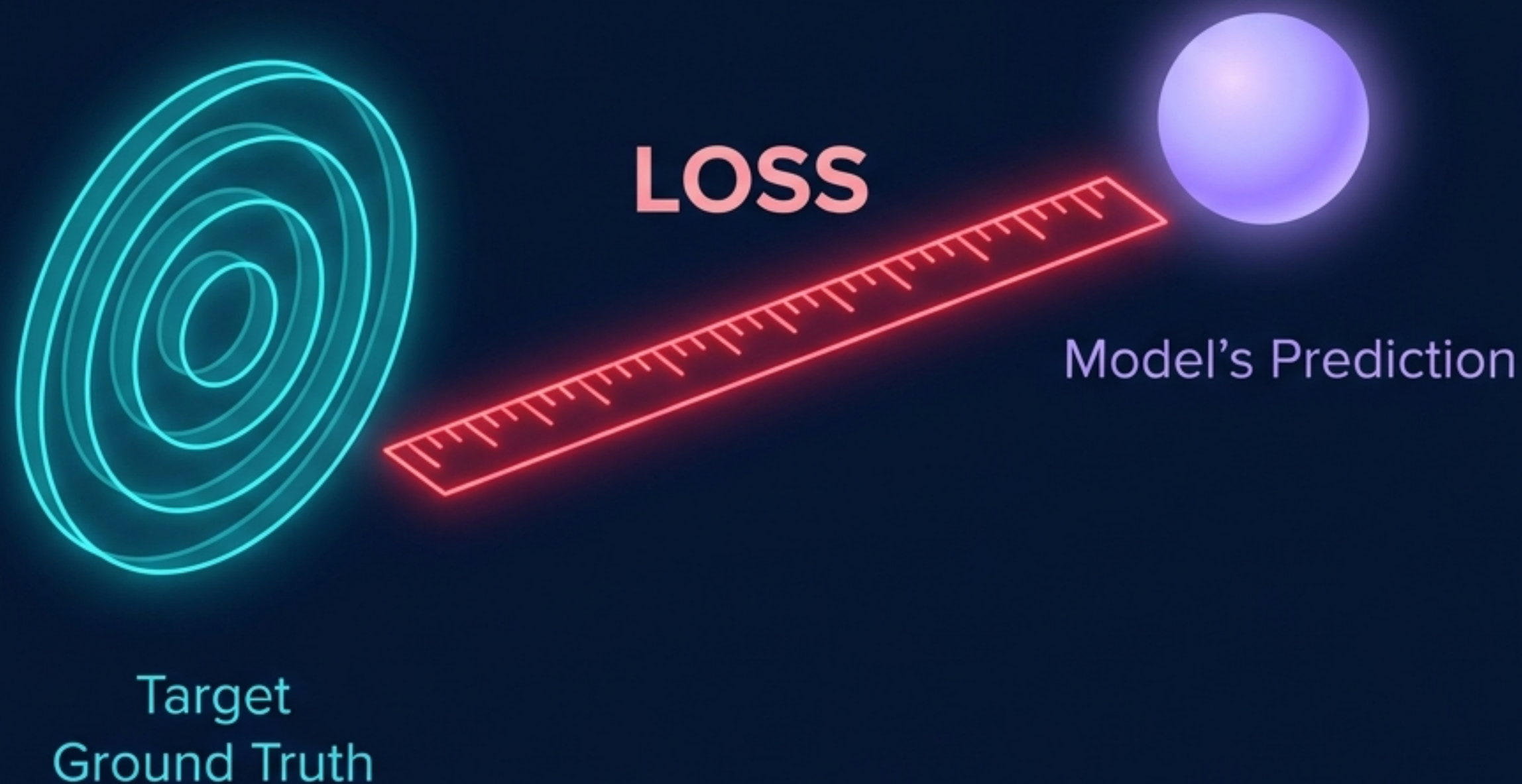


Interactive Challenge – The Robotic Assembly Line

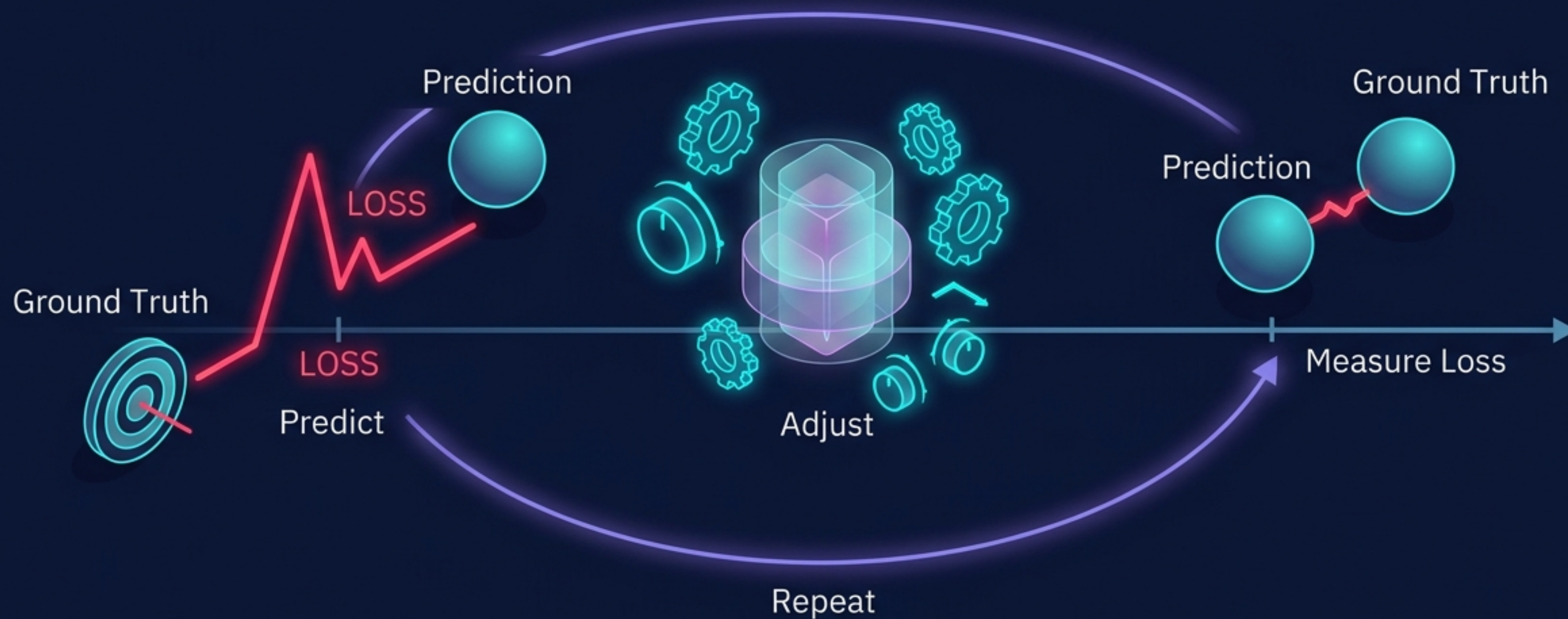


Which are Classification and which is Regression?

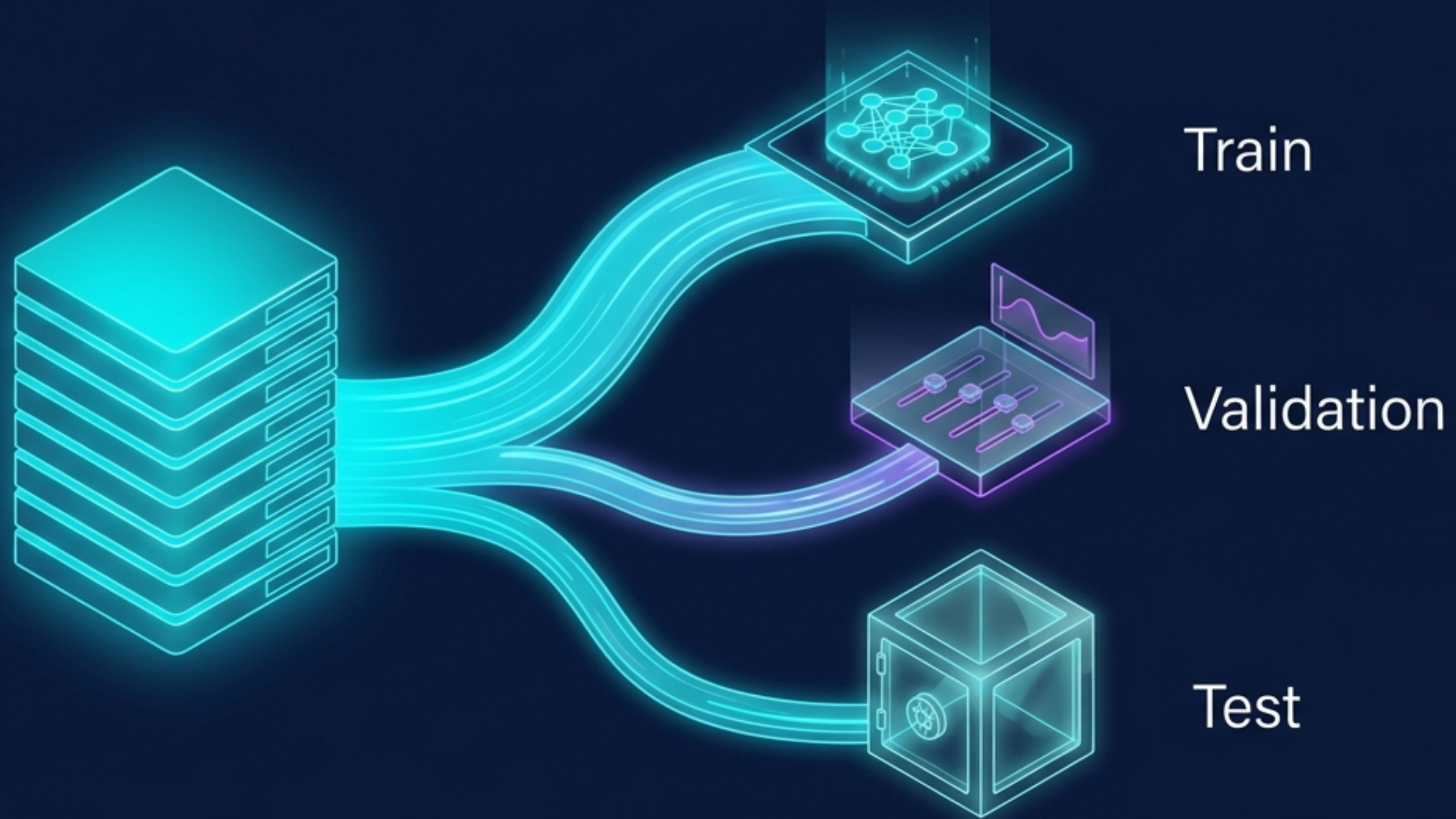
The Loss Function – Measuring the Gap



Training – Reducing Loss Over Time

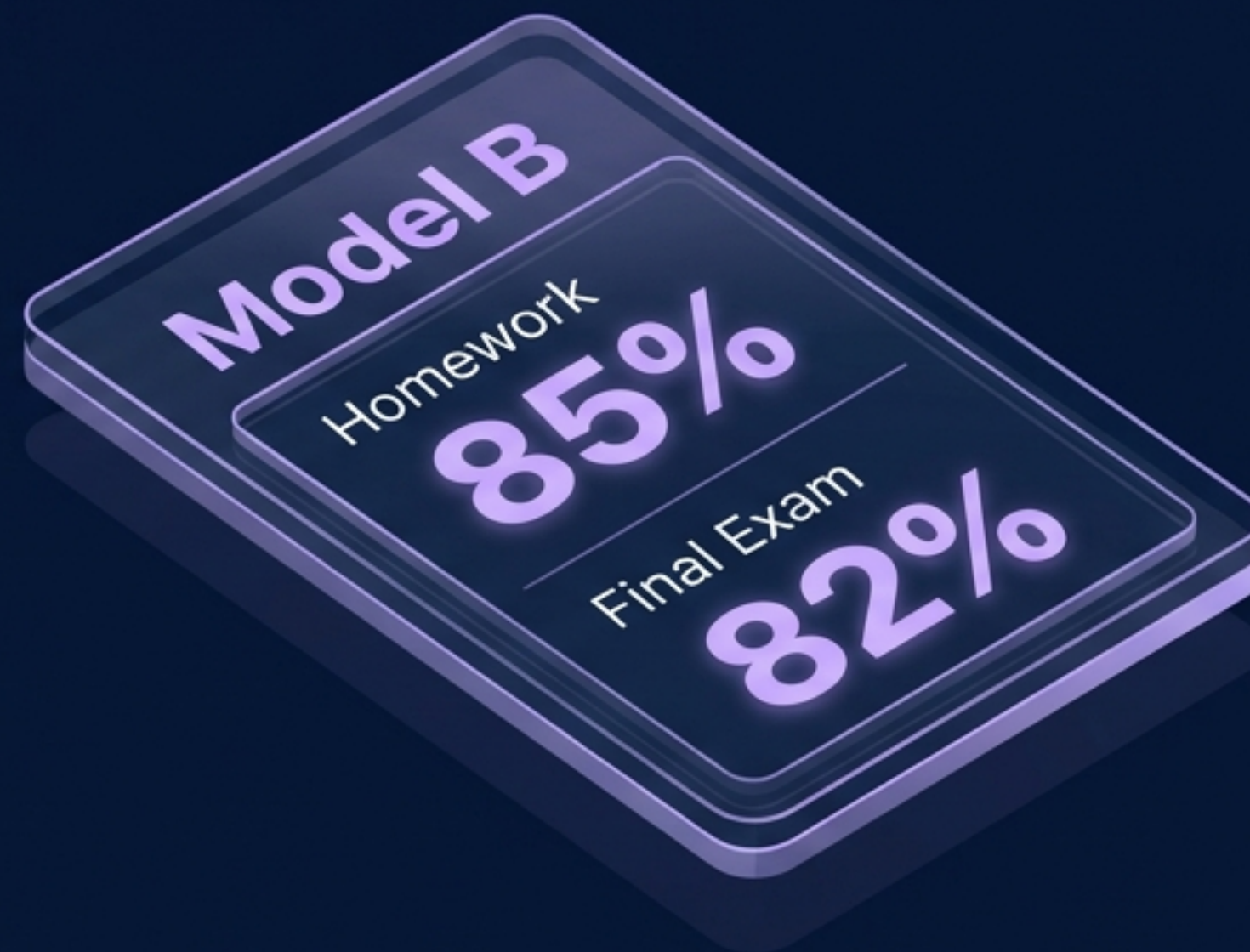


Train, Validation, and Test Split



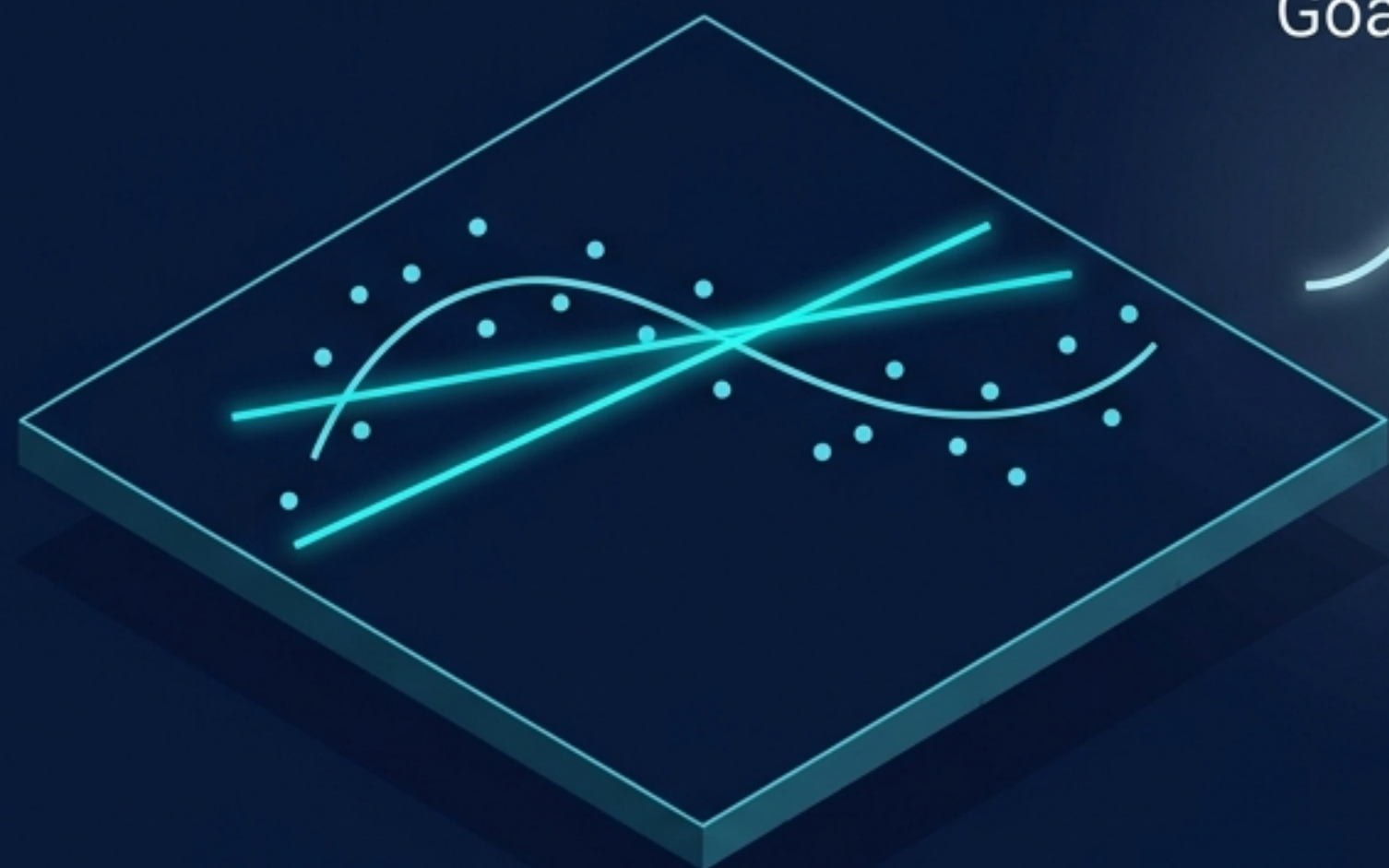
Which model would you trust?

Who learned, and who memorized?



Underfitting vs Overfitting

Underfitting

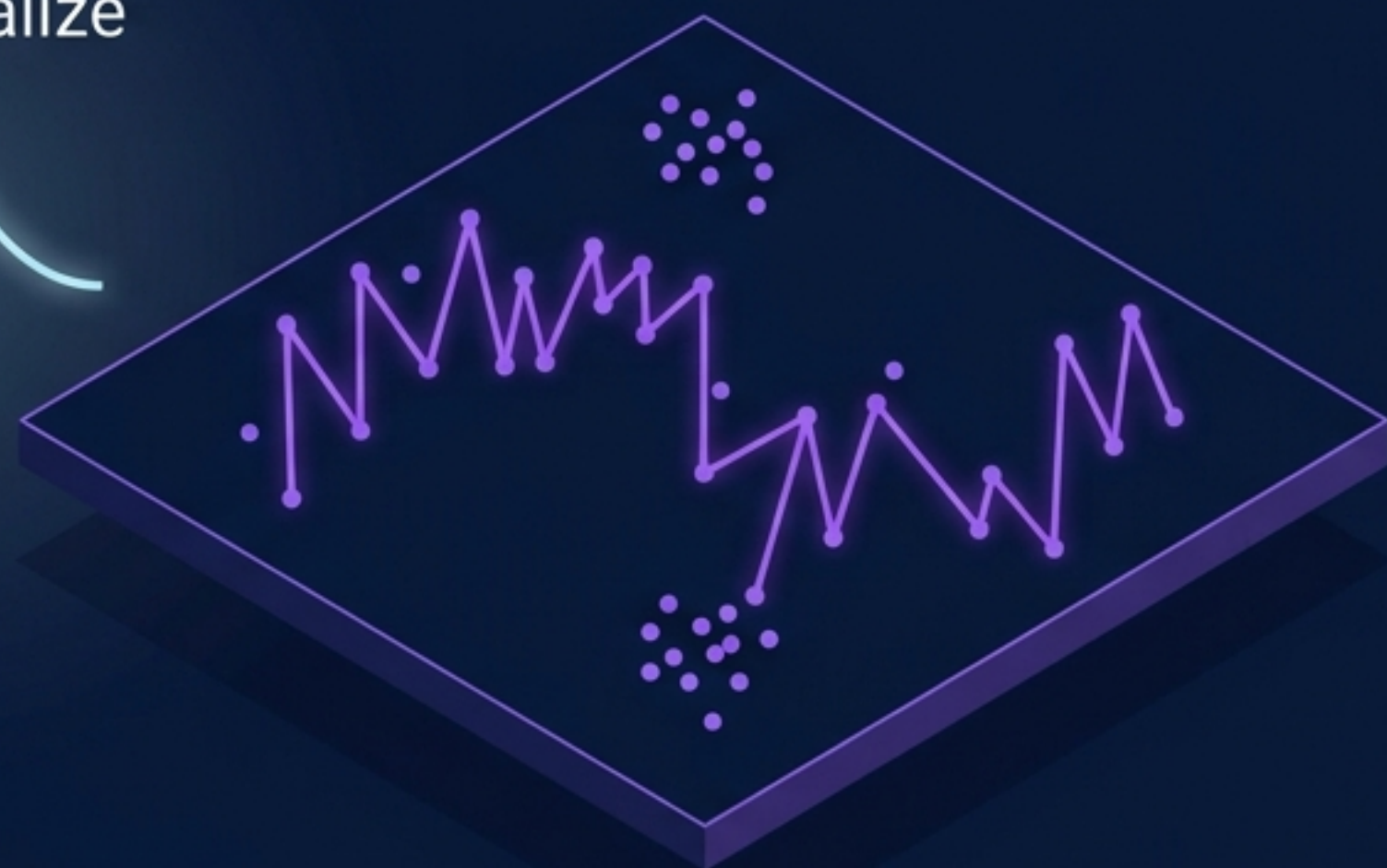


Too Simple

Goal: Generalize

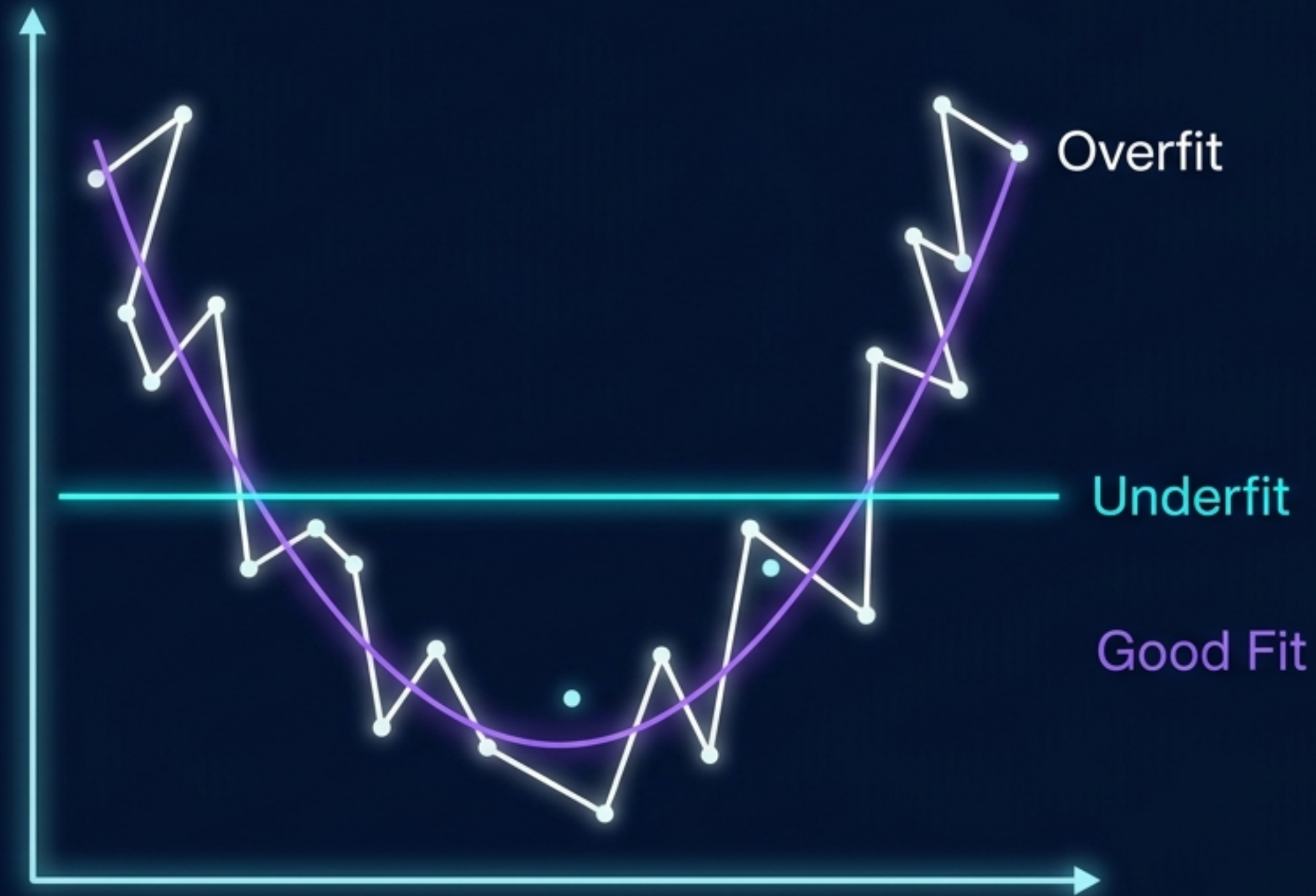


Overfitting



Too Specific

Visualizing the Fit



Interactive Challenge: The Diagnosis



Which model is too lazy, which is too obsessed, and which is just right?

Bias vs. Variance



High Bias

🎯 Underfitting



Low Bias &
Low Variance



High Variance

🧠 Overfitting

The Big Picture – Lesson Summary

